

Solution Requirements

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users securely access the application using login credentials before making predictions</i>	High
2	Authorization levels	<i>Admin manages datasets and mode updates. Users can only enter Weather parameters and view prediction results</i>	High
3	External interfaces	<i>Integrates with weather/Rainfall datasets and optionally weather APIs for real-time data</i>	High

4	Transactions processing	<i>Accept weather parameters, process them through the ML model, and generate flood prediction results instantly.</i>	High
5	Reporting	<i>Display prediction results, flood risk level, and weather information on a dashboard.</i>	Medium
6	Business rules, etc.	<i>Prediction is generated only when all required input fields are valid. Invalid or missing values are rejected.</i>	High
7	Compliance to laws or regulations	<i>Protect user data and follow standard data privacy and security practices.</i>	Medium
8	<i>Other</i>	<i>Provide recommendations and safety alerts based on predicted flood risk.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>Prediction should be generated quickly after input submission.</i>	<i>Response time < 3 seconds</i>
2	Scalability	<i>System should support increasing users and larger datasets.</i>	<i>Supports 500+ concurrent users</i>
3	Security & Data Privacy	<i>Secure user inputs and protect stored data from unauthorized access.</i>	<i>HTTPS encryption and secure authentication</i>
4	Reliability & Availability	<i>Application should remain available with minimal downtime.</i>	<i>99% uptime</i>
5	Usability & Accessibility	<i>Interface should be simple, responsive, and easy for all users.</i>	<i>Prediction in less than 3 clicks</i>
6	Other	<i>Application should be maintainable and easy to update with new ML models.</i>	Modular and documented code